

delivers outstanding output power, superior spectral purity, exceptionally low close-in phase noise, and minimal wideband noise, making it ideal for analogue microwave signal generation.

The base model covers frequencies from DC to 2.025GHz, ideal for general lab and educational use. Advanced models extend capabilities for radar and telecommunications applications, reaching frequencies up to 4.05GHz and 6.075GHz. Optionally, high-end models can utilise a frequency doubler to achieve frequencies up to 8.10GHz, accommodating complex systems. With these enhancements, the latest signal generators support a wide range of applications, showcasing adaptability and driving modern wireless innovations in telecommunications and radar technology.

Performance and control

New signal generators set new standards for signal quality and integrity, offering low error vector magnitude (EVM) and superior adjacent channel power ratio (ACPR). These features ensure the reliability and clarity essential for high stakes testing across multiple industries. With exceptionally low phase noise and the integration of direct digital synthesis (DDS) and custom digital-to-analogue converter (DAC) technology, these generators deliver unparalleled signal fidelity, minimising distortion and maintaining high performance.

Madhukar Tripathi, Associate Director - Business Development & Marketing at Anritsu India Pvt Ltd, informed about the launch of their MG36271A signal generator, designed for high-frequency applications. It delivers phase noise of -136dBc/Hz (typical) and -140dBc/Hz (measured) at 10GHz and 10kHz offset, with low harmonics for exceptional signal purity. With +9dBm output power at 70GHz, this generator meets Q/V

band measurement requirements. It offers extensive modulation capabilities and superior frequency stability, making it ideal for aerospace, defence, communications, and research applications.

Rigol DG1022Z signal generators enhance design and portability by introducing USB power and control, eliminating the need for external power supplies. Engineers can fine-tune signals with frequency adjustments down to 100Hz and power increments of 0.5dB. The device supports programmable linear frequency sweeps and fast frequency switching (up to 100 microseconds), making it versatile for complex experiments and dynamic environments.

Connectivity for demanding fields

Designed for versatility, these devices feature dual front-panel outputs covering overlapping frequency ranges, ideal for aerospace and defence applications. Their low phase noise ensures pristine signal quality, critical in high-stakes environments.

The 2024 models of the Rohde & Schwarz SMW200A, Anritsu MG3710A, and Tektronix TSG4100A offer dual outputs with low phase noise and optional enhancements like frequency doublers and differential clock outputs. Their comprehensive connectivity options (GPIO, RS-232, and Ethernet) make them ideal for automated and remote operations across diverse industries.

Adding to their robust functionality, these models offer optional enhancements, such as output frequency doublers and differential clock outputs. These features facilitate precise timing and synchronisation necessary for managing complex setups in specialised testing scenarios. They are equipped with comprehensive connectivity options, including GPIO, RS-232, and Ethernet. This array of connections

not only supports seamless integration into automated systems but also enables efficient remote operation capabilities. Such extensive connectivity ensures that these devices can be easily incorporated into diverse operational frameworks, enhancing their utility and flexibility in various technological landscapes.

The trajectory of signal generators within the realms of telecommunications, defense, aerospace, and automotive industries signifies a vibrant era of innovation and growth. The integration of cutting-edge technologies like IoT, 5G, and AI into these devices is revolutionising the way electronic devices are tested and maintained. The emergence of advanced models featuring super-high-frequency signal generators and compact multichannel vector signal generators underscores a shift towards more efficient, precise, and space-conserving testing setups. Leading companies are setting new benchmarks in signal generation that promise to enhance productivity and adaptability across various high-stakes industries. This ongoing evolution not only supports current technological demands but also lays the groundwork for future advancements, ensuring that signal generators remain indispensable tools in the development and maintenance of sophisticated electronic systems. **EFY**

Akanksha Sondhi Gaur is Senior Technology Journalist at EFY. She has a German patent and brings a robust blend of seven years of industrial and academic prowess to the table. Passionate about electronics, she has penned numerous research papers showcasing her expertise and keen insight.

Nidhi Agarwal is also Senior Technology Journalist at EFY. She is an Electronics and Communication Engineer with over five years of academic experience. Her expertise lies in working with development boards and IoT cloud. She enjoys writing as it enables her to share her knowledge and insights related to electronics with like-minded techies.